

Speaker Cues for Proposal Defense Presentation

Slide 1: Title Page

Short Cue: Greeting + thesis goal + presentation overview

Detailed Description:

Good afternoon professors, I am Yujie Feng. The goal of my thesis is to build a SaaS-oriented AI inspection platform for power grids that can detect various types of equipment damage and anomalies. Today I will walk you through three main parts: first, the platform vision and architecture; second, the end-to-end proof of concept we've completed (Module A: bird-nest detection); and third, the roadmap for platformization, including plans for oil leak detection and infrared high-temperature detection modules.

Slide 2: Project Vision: A Scalable SaaS Inspection Platform

Short Cue: Emphasize SaaS architecture, not just a single model; explain three-column structure

Detailed Description:

The key point here is that this thesis targets the entire SaaS architecture that can onboard multiple AI capabilities for different grid bureaus, not just a single detection model. Let me break down the three pillars:

Left column - Multi-modal Inputs: The platform accepts diverse data sources: UAV RGB imagery, infrared imagery, sensor telemetry streams, and historical inspection records. This multi-modal approach is essential for comprehensive equipment monitoring.

Center column - AI Platform Core: This is the heart of the system. The SaaS backend provides unified authentication with role-based access control, multi-tenant isolation for different grid bureaus, and infrastructure like task queues and data lakes. The AI model registry is where we plug in different detection modules. Currently, we have Module A (bird nest detection) as our PoC, with Module B (oil leak detection) and Module C (infrared high-temperature detection) planned.

Right column - Enterprise Outputs: The platform delivers actionable outputs: an alert center and dashboard for real-time monitoring, maintenance work orders for field teams, a visual analytics portal for analysis, and APIs for mobile integration.

This architecture ensures scalability and extensibility—we can add new detection modules without rebuilding the entire system.

Slide 3: Accomplished (1): End-to-End PoC -- Module A

Short Cue: Why bird nests? Minimal vertical slice proves feasibility

Detailed Description:

We chose bird-nest detection as our first module because it's a high-frequency, high-risk issue with clear visual patterns, making it ideal to validate the entire pipeline from data to deployment.

The flow chart shows our complete vertical slice: We started with a small dataset of 43 training and 11 validation images. We fine-tuned a Mask R-CNN model, applied engineering optimizations like DataLoader parallelism, mixed precision training, and learning rate schedulers. Finally, we deployed it as a Flask microservice. This end-to-end implementation proves that the pipeline works and can be replicated for other modules.

Slide 4: PoC Progress (1): Training Results & Metrics

Short Cue: Present COCO metrics table; highlight AP50=1.0 and strong overall AP scores

Detailed Description:

Let me present the quantitative results. On the left, we have the COCO validation metrics. For bounding box detection, we achieved an AP@[0.50:0.95] of 0.805, with AP50 at 1.000 and AP75 at 1.000. For segmentation, we achieved 0.795 AP@[0.50:0.95], with AP50 at 1.000 and AP75 at 0.919. The average recall is 0.837 for bbox and 0.828 for segmentation.

On the right, I've listed the training configuration. We used Mask R-CNN with ResNet50 backbone and Feature Pyramid Network v2, pre-trained on COCO. The dataset is small—only 54 images total—but we used aggressive data augmentation. Input size was 512 by 512 pixels, batch size 4. We used AdamW optimizer with a learning rate of 5 times 10 to the negative 4, with OneCycleLR scheduler. We trained for 40 epochs with early stopping capability.

The AP50 of 1.0 means that at IoU threshold 0.5, we achieved perfect precision and recall. However, as I'll discuss later, this is partly because our validation set mirrors the training distribution.

Slide 5: PoC Progress (2): Training Dynamics

Short Cue: Point to loss curves; mention convergence and GPU optimization

Detailed Description:

The loss curves show healthy convergence. Training loss dropped from 1.41 to 0.14, while validation loss decreased from 0.65 to 0.19. The model plateaus around epoch 20, indicating that further training would yield diminishing returns.

A key engineering achievement was optimizing GPU utilization. Initially, we had low GPU usage due to data loading bottlenecks. After implementing DataLoader parallelism with persistent workers, enabling cuDNN benchmark mode, and adjusting batch size and augmentation parameters, we stabilized GPU utilization at 70 to 85 percent. This reduced training time to about 27 seconds per epoch, making iterative experimentation much more efficient.

Slide 6: PoC Progress (3): Precision-Recall Analysis

Short Cue: Compare PR curves at IoU 0.50 / 0.75 / 0.90

Detailed Description:

The updated plots overlay three IoU thresholds. At IoU = 0.50 the precision curve remains almost flat at 1.0, confirming negligible false positives. Increasing the requirement to IoU = 0.75 lowers the plateau to roughly 0.93, which aligns with the AP75 score from COCO evaluation. At a very strict IoU = 0.90 the precision curve dips toward ~0.70 as recall grows, highlighting that localization begins to slip when we demand near-perfect overlap. This contrast illustrates that while coarse localization is rock-solid, there is still room to improve mask and box refinement.

Slide 7: PoC Progress (4): IoU Distributions & Threshold Analysis

Short Cue: Right-skewed IoU = strong localization; threshold 0.4-0.5 is optimal

Detailed Description:

The bounding box IoU histogram shows a right-skewed distribution, with most detections having IoU above 0.8. This indicates strong localization accuracy—the model not only detects bird nests but also localizes them precisely.

The threshold sweep analysis on the right shows how precision, recall, and F1 score vary with the detection confidence threshold, evaluated at $\text{IoU} \geq 0.5$. Given that $\text{AP50}=1.0$, precision remains consistently high (close to 1.0) across a wide range of thresholds. At very low thresholds (below 0.3), we get high recall but may have some false positives. The F1-optimal point is around 0.4 to 0.5, which balances recall and precision. However, higher thresholds (0.6-0.8) are also viable—they maintain excellent precision (near 1.0) with only moderate recall reduction, making them suitable for applications where minimizing false positives is critical. We use 0.4-0.5 in the deployed service as a balanced default, but users can adjust based on their specific precision-recall trade-off requirements.

Slide 8: PoC Progress (5): Production-Ready Microservice

Short Cue: Production-ready Flask API with batch processing, visualization, and enterprise features

Detailed Description:

The trained model has been deployed as a production-ready Flask REST API microservice with comprehensive features beyond basic inference.

Core API Features: The service supports batch image upload, allowing operators to process multiple images simultaneously—essential for field inspection workflows. It provides real-time inference with visual annotations, showing both bounding boxes and segmentation masks overlaid on the original images. The interactive web interface enables users to adjust confidence thresholds dynamically and view results immediately.

Enterprise Capabilities: Beyond basic detection, the service includes statistical dashboards showing total detections, average confidence scores, and distribution charts. Users can download individual

annotated images or batch download all results as a ZIP file. The system is memory-optimized to handle large images automatically, with intelligent resizing and compression to prevent memory errors.

Platform Significance: This deployment demonstrates "model as a service"—the core concept of our SaaS platform. It serves as the first entry in our future AI model registry, showing how new modules can be integrated. The production-ready error handling, memory management, and user-friendly interface validate that the platform can scale to real-world operations. It enables integration with alert centers and maintenance systems, completing the end-to-end workflow from detection to action.

Visual Demonstrations: The left screenshot shows single image detection with detailed annotations—bounding boxes, segmentation masks, and confidence scores displayed in a clear table format. The right screenshot demonstrates the real-time inference and visualization capabilities, including batch processing results, statistical summaries with confidence distribution charts, and the interactive dashboard that provides comprehensive insights at a glance. These visualizations showcase the production-ready interface that will be essential when we scale to multiple grid bureaus.

Slide 9: Key Insight: Generalization is the Core Challenge

Short Cue: Perfect scores are misleading; real challenge is generalization to new conditions

Detailed Description:

The most valuable outcome of Module A is not the high AP scores, but the realization that generalization—not raw accuracy—is the real hurdle for a SaaS platform.

The Trap of a "Perfect" Score: While we achieved AP50 of 1.0 and IoU near 1.0, this is largely because our validation data mirrors the training scenes—similar lighting, weather conditions, and equipment types. In real operations, when we encounter fog, backlight, different camera angles, or new structure types, accuracy might drop sharply. This is the generalization gap that every production AI system must address.

Platform Lesson: This insight applies to all modules—whether it's bird nest detection, oil leak detection, or infrared high-temperature detection. Each module must solve the generalization problem to new conditions. This means data augmentation, negative sampling to reduce false positives, and multi-scene evaluation are not optional features—they are platform-level requirements that must be built into the infrastructure from the start.

Slide 10: Next Steps: Build Horizontally, Expand Vertically

Short Cue: Two-track approach: SaaS infrastructure (Track 1) + new modules (Track 2)

Detailed Description:

Moving forward, I will evolve the SaaS backbone while concurrently delivering the next AI modules. This two-track approach ensures we build infrastructure and validate extensibility simultaneously.

Track 1—SaaS Platform (Weeks 5-10): The goal is to establish core infrastructure and standards. Tasks include: First, implementing a multi-tenant backend with role-based access control and task orchestration. This will allow different grid bureaus to use the platform securely and independently. Second, defining APIs for the AI model registry so new modules can be plugged in consistently. Third, productizing the PoC training utilities—data augmentation pipelines, logging systems, and evaluation frameworks—so they can be reused across modules.

Track 2—New Modules (Weeks 11-18): The goal is to validate extensibility. Tasks include: First, kicking off either Module B (oil leak detection) or Module C (infrared high-temperature detection). Second, reusing platform capabilities for data processing and training, proving that the infrastructure we built in Track 1 actually works. Third, onboarding new modules into the registry and microservice catalog, demonstrating that the platform can scale horizontally.

Slide 11: Work Plan and Schedule

Short Cue: Point to Gantt chart; explain two-track timeline

Detailed Description:

The Gantt chart visualizes the 20-week timeline. The green bars represent weeks 1-4, which we've completed—this was the PoC implementation for Module A.

The blue bars show the future work. Weeks 5-10 are dedicated to Track 1, building the SaaS platform infrastructure. Weeks 11-15 focus on Track 2a, implementing Module B for oil leak detection. Weeks 16-18 cover Track 2b, implementing Module C for infrared high-temperature detection. Finally, weeks 19-20 are reserved for final integration, testing, and report writing.

The two-track approach allows us to build infrastructure and add new capabilities in parallel, maximizing progress while ensuring the platform remains extensible and maintainable.

Slide 12: Discussion Points

Short Cue: Present three strategic questions; invite feedback

Detailed Description:

I'd like to invite your guidance on three strategic questions that will shape the next phase of the project:

Question 1—SaaS Backend Depth: Should I prioritize feature completeness first by building a prototype RBAC system that demonstrates the concept, or should I invest in production-grade authentication and authorization from day one? The trade-off is between speed of iteration and long-term maintainability.

Question 2—Module B Technical Choices: For oil leak detection, the visual characteristics might be different from bird nests. Should I consider different architectures like semantic segmentation with U-Net, which might be better for detecting continuous regions, or should I stick with instance segmentation (Mask R-CNN) for consistency across modules? The trade-off is between model specialization and platform uniformity.

Question 3—Module C Integration: For infrared high-temperature detection, should it be a standalone module that processes infrared images independently, or should it be fused with RGB-based detection for multi-modal analysis? The trade-off is between modularity and the potential performance gains from multi-modal fusion.

I'm looking forward to your insights on these architectural decisions. Thank you for your attention.